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Inspect Rich Documents with Gemini Multimodality and Multimodal RAG

Course · 4 hours 75% 45 minutes complete

🏠 Course overview

Inspect Rich Documents with Gemini Multimodality and Multimodal RAG

✓ 🔗 Multimodality with Gemini

✓ 🔗 Using Gemini for Multimodal Retail Recommendations

✓ 🔗 Multimodal Retrieval Augmented Generation (RAG) using the Gemini API in Vertex AI

Inspect Rich Documents with Gemini

○ 🔗 Multimodality and Multimodal RAG:

Quick tip: Review the prerequisites before you run the lab

Start Lab

01:00:00

Multimodality with Gemini

🔗 Lab ⌚ 1 hour 💰 No cost ✓ Intermediate

★★★★★ Rate Lab

📘 This lab may incorporate AI tools to support your learning.

GSP1210



Lab instructions and tasks —/100

GSP1210

Overview

Objectives

Setup and requirements

Task 1. Open the notebook in Vertex AI Workbench

Task 2. Set up the notebook

Task 3. Use the Gemini Flash model

Congratulations!

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Overview

This lab introduces you to [Gemini](#), a family of multimodal generative AI models developed by Google. You use the Gemini API to explore how Gemini Flash can understand and generate responses based on text, images, and video.

Gemini’s multimodal capabilities enable it to:

- **Analyze images:** Detect objects, understand user interfaces, interpret diagrams, and compare visual similarities and differences.
- **Process videos:** Generate descriptions, extract tags and highlights, and answer questions about video content.

You experiment with these features through hands-on tasks using the Gemini API in Vertex AI.

Prerequisites

Before starting this lab, you should be familiar with:

- Basic Python programming.

Objectives

In this lab, you:

- Interact with the Gemini API in Vertex AI.
- Use the Gemini Flash model to analyze images and videos.
- Provide Gemini with text, image, and video prompts to generate informative responses.
- Explore practical applications of Gemini’s multimodal capabilities.

Before you click the Start Lab button

Read these instructions. Labs are timed and you cannot pause them. The timer, which starts when you click **Start Lab**, shows how long Google Cloud resources are made available to you.

This hands-on lab lets you do the lab activities in a real cloud environment, not in a simulation or demo environment. It does so by giving you new, temporary credentials you use to sign in and access Google Cloud for the duration of the lab.

To complete this lab, you need:

- Access to a standard internet browser (Chrome browser recommended).

Note: Use an Incognito (recommended) or private browser window to run this lab. This prevents conflicts between your personal account and the student account, which may cause extra charges incurred to your personal account.

- Time to complete the lab—remember, once you start, you cannot pause a lab.

How to start your lab and sign in to the Google Cloud console

1. Click the **Start Lab** button. If you need to pay for the lab, a dialog opens for you to select your payment method. On the left is the Lab Details pane with the following:

- The Open Google Cloud console button
- Time remaining
- The temporary credentials that you must use for this lab
- Other information, if needed, to step through this lab

2. Click **Open Google Cloud console** (or right-click and select **Open Link in Incognito Window** if you are running the Chrome browser).

The lab spins up resources, and then opens another tab that shows the Sign in

Tip: Arrange the tabs in separate windows, side-by-side.

Note: If you see the **Choose an account** dialog, click **Use Another Account**.

3. If necessary, copy the **Username** below and paste it into the **Sign in** dialog.

"Username"



You can also find the Username in the Lab Details pane.

4. Click **Next**.

5. Copy the **Password** below and paste it into the **Welcome** dialog.

"Password"



You can also find the Password in the Lab Details pane.

6. Click **Next**.

your Google Cloud account credentials.

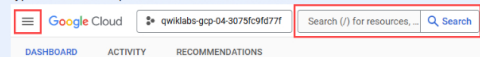
Note: Using your own Google Cloud account for this lab may incur extra charges.

7. Click through the subsequent pages:

- Accept the terms and conditions.
- Do not add recovery options or two-factor authentication (because this is a temporary account).
- Do not sign up for free trials.

After a few moments, the Google Cloud console opens in this tab.

Note: To access Google Cloud products and services, click the **Navigation menu** or type the service or product name in the **Search** field.



Task 1. Open the notebook in Vertex AI Workbench

1. In the Google Cloud console, on the **Navigation menu** () click **Vertex AI > Workbench**.
2. Find the **Workbench instance name** instance and click on the **Open JupyterLab** button.

The JupyterLab interface for your Workbench instance opens in a new browser tab.

1. Open the **notebook name** file.
2. In the **Select Kernel** dialog, choose **Python 3** from the list of available kernels.
3. Run through the **Getting Started** and the **Import libraries** sections of the notebook.
 - For **Project ID**, use **Project ID**, and for **Location**, use **Region**.

Note: You can skip any notebook cells that are noted *Colab only*. If you experience a 429 response from any of the notebook cell executions, wait 1 minute before running the cell again to proceed.

Task 3. Use the Gemini Flash model

text, image(s), and video in your prompt requests and get text or voice responses.

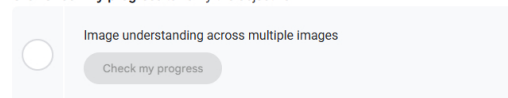
In this task, run through the specified notebook cells to see how to use the Gemini Flash model. Return here to check your progress as you complete the objectives.

Image understanding across multiple images

One of the Gemini's capabilities is being able to reason across multiple images. In this example, you use Gemini to calculate the total cost of groceries using an image of fruits and a price list.

Run through the **Image understanding across multiple images** section of the notebook.

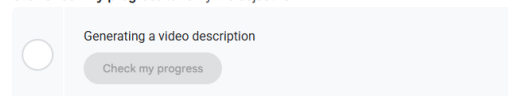
Click **Check my progress** to verify the objective.



Gemini can also extract tags throughout a video and retrieve extra information beyond the video contents. In this example, you use Gemini to extract tags and retrieve extra information from different videos:

Run through the **Generating a video description** section of the notebook.

Click [Check my progress](#) to verify the objective.



Audio understanding

Gemini can directly process audio for long-context understanding. In this example, you use Gemini to process audio for long-context understanding:

Run through the **Audio understanding** section of the notebook.

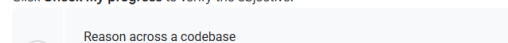
Click **Check my progress** to verify the objective.

Reason across a codebase

Gemini can directly process audio for long-context understanding. In this example, you use Gemini to process audio for long-context understanding:

Run through the **Reason across a codebase** section of the notebook.

Click **Check my progress** to verify the objective.





Check my progress

Video and audio understanding

Run through the **Video and audio understanding** section of the notebook.

Click **Check my progress** to verify the objective.



Video and audio understanding
Check my progress

All modalities (images, video, audio, text) at once

Gemini is natively multimodal and supports interleaving of data from different modalities. In this example, you try a mix of audio, visual, text, and code inputs in the same input sequence.

Run through the **All modalities (images, video, audio, text) at once** section of the notebook.

Click **Check my progress** to verify the objective.

All modalities (images, video, audio, text) at once

Generating recommendations based on provided images

Gemini is capable of image comparison and providing recommendations. This is particularly useful for retail companies who want to provide users product recommendations based on their current setup.

Run through the **Generating recommendations based on provided images** section of the notebook.

Click **Check my progress** to verify the objective.



Generating recommendations based on provided images
Check my progress

Understand entity relationships in technical diagrams

Advanced steps, such as optimization or code generation. In this example, you see how Gemini can decipher an entity relationship (ER) diagram, understand the relationships between tables, identify requirements for optimization in a specific environment like BigQuery, and even generate corresponding code.

Run through the **Understand entity relationships in technical diagrams** section of the notebook.

Click **Check my progress** to verify the objective.



Understand entity relationships in technical diagrams
Check my progress

Compare images for similarities and differences

Gemini can compare images and identify similarities or differences between objects. In this example, you use Gemini to compare two images of the same location and identify the differences between them.

Run through the **Compare images for similarities and differences** section of the

Click **Check my progress** to verify the objective.



Compare images for similarities and differences
Check my progress

Congratulations!

You have now completed the lab! In this lab, you learned how to use the Gemini API in

Vertex AI to generate text from text and image(s) prompts.

Next steps / learn more

Check out the following resources to learn more about Gemini:

- [Generative AI on Vertex AI Documentation](#)
- [Generative AI on YouTube](#)
- Explore the Vertex AI [Cookbook](#) for a curated, searchable gallery of notebooks for Generative AI.
- Explore other notebooks and samples in the [Google Cloud Generative AI repository](#).

Google Cloud training and certification

...helps you make the most of Google Cloud technologies. [Our classes](#) include technical skills and best practices to help you get up to speed quickly and continue your learning journey. We offer fundamental to advanced level training, with on-demand, live, and virtual options to suit your busy schedule. [Certifications](#) help you validate and prove your skill and expertise in Google Cloud technologies.

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